

# A4. Emergence of Segregation in Evolving Social Networks: A Link-Rewiring Generalization of Schelling's Model

Omar Morales Ortega  
Complex Networks

May 28, 2026

## 1 Introduction

In 1971, Thomas Schelling introduced a famous agent-based model to explain how racial segregation occurs in cities [2]. His main finding was striking: even if individuals are relatively tolerant and only prefer a small fraction of similar neighbors, their collective, decentralized choices inevitably lead to a completely segregated society. This model is a classic example of an emergent phenomenon, showing how micro-level individual preferences create macro-level structural patterns that nobody explicitly intended.

However, the traditional Schelling model is limited by its spatial layout, which relies on a rigid two-dimensional grid or lattice. While a grid works well for mapping physical neighborhoods, it does not capture modern social interactions. Human relationships are better represented as complex networks where connectivity is fluid, irregular, and changes over time, rather than being restricted to immediate physical neighbors.

To adapt Schelling's ideas to network science, this report focuses on a "link-rewiring" approach based on the work of Henry, Pralat, and Zhang (2011) [1]. Instead of moving agents to different empty nodes on a fixed graph, this model allows the network's edges to evolve. The key innovation is that agents rewire their connections based on their personal satisfaction: an agent must maintain a minimum fraction of same-type neighbors to be satisfied. Unsatisfied agents systematically remove cross-group connections and establish new ties until equilibrium is reached.

The objective of this assignment is to implement this rewiring model in Python and analyze how segregation develops over time. Following the task requirements, we will explore the key parameters of the system, use sequential network visualizations to show the clustering process, and perform ensemble averaging over multiple simulation runs to guarantee statistically reliable results.

## 2 Model Generalization and Mechanism

### 2.1 Limitations of the Lattice Approach

The classic Schelling model places agents on a rigid two-dimensional grid where each agent can only interact with their immediate Moore neighborhood (the 8 surrounding cells). While this spatial structure is highly intuitive for geographical neighborhoods, it has significant limitations when applied to broader social dynamics.

First, a grid assumes a perfectly uniform distribution of contacts, meaning every single individual has the exact same potential capacity for relationships. In reality, human social structures are highly irregular; some individuals are "hubs" with many connections, while others

are relatively isolated. Second, spatial grids imply that physical proximity is the only driver of relationships. In a modern context, social ties are fluid and not strictly bound by geography. Moving a physical agent to a new location change all their neighbors simultaneously, which is a very drastic structural shift.

## 2.2 The Link-Rewiring Framework

To address these limitations, we implement the generalization proposed by Henry, Prałat, and Zhang (2011) [1], which shifts the focus from spatial movement to network adaptation. Instead of moving nodes across a static canvas, the agents remain fixed as nodes in a graph  $G = (V, E)$ , and the connections (edges) between them dynamically evolve over time.

We consider a population of  $N$  nodes split into two distinct groups (Group 0 and Group 1) based on a population ratio  $\rho$ . The total number of edges  $|E|$  in the network is kept strictly constant throughout the simulation to maintain a stable network density.

**Agent Satisfaction:** At any point in time, an agent (node)  $u$  measures its personal satisfaction as the fraction of its neighbors who belong to the same group as itself:

$$S(u) = \frac{|\{v \in N(u) : \text{type}(v) = \text{type}(u)\}|}{|N(u)|} \quad (1)$$

where  $N(u)$  denotes the set of neighbors of node  $u$ . A node is considered **satisfied** if  $S(u) \geq \tau$ , where  $\tau$  is a satisfaction threshold parameter (in our experiments,  $\tau = p_{\text{aversion}}$ ). A node is **unsatisfied** if  $S(u) < \tau$ .

The discrete-time dynamics of the model follow a strict algorithmic loop at each time step:

1. **Satisfaction Assessment:** The system identifies all unsatisfied nodes—those whose same-type neighbor fraction falls below the threshold  $\tau$ . If no unsatisfied nodes exist, the system has reached a steady state and the simulation terminates.
2. **Synchronous Rewiring:** In each step, all unsatisfied nodes execute the following rewiring procedure. For each unsatisfied node  $u$ :
  - (a) **Cross-Edge Identification:** Node  $u$  identifies all its neighbors from the opposite group (cross-neighbors).
  - (b) **Edge Removal:** One cross-neighbor  $v$  is selected uniformly at random from this pool, and the edge  $(u, v)$  is deleted from the network.
  - (c) **Random Rewiring:** To maintain a constant edge count, node  $u$  randomly selects a node  $w$  from the entire network that is not already a neighbor:
    - The only constraint is that  $(u, w) \notin E$  and  $w \neq u$ .
    - Crucially,  $w$  can belong to **either group**—there is no restriction on type.

A new edge is established:  $E \leftarrow E \cup \{(u, w)\}$ .
3. **Post-Rewiring Evaluation:** After all unsatisfied nodes have completed their simultaneous rewiring, the satisfaction levels of all affected nodes are re-evaluated in the next iteration. This may result in previously satisfied nodes becoming unsatisfied (or vice versa), driving the dynamics forward.

This synchronous, group-based rewiring mechanism closely mirrors the original Schelling model: when an agent is dissatisfied with its current neighborhood, it moves to a randomly available location (here, a random unoccupied edge slot) without predetermined control over what types of neighbors it will have at that new location.

### 2.3 Quantifying Segregation

To track the emergence of segregation at any point during the simulation, we calculate the global Homophily Index ( $H$ ). This metric tracks the total fraction of edges that connect nodes belonging to the exact same group:

$$H = \frac{E_{\text{internal}}}{E_{\text{total}}} = \frac{|\{(u, v) \in E : \text{type}(u) = \text{type}(v)\}|}{|E|} \quad (2)$$

When the network is initialized as a completely random graph, the homophily index floats around  $H \approx 0.5$  (assuming a 50/50 group split).

## 3 Experimental Setup and System Parameters

The current experiments are implemented in the notebook using a synchronous, satisfaction-driven edge rewiring algorithm. Each simulation begins with  $N = 1000$  nodes assigned to two groups according to a group ratio  $\rho$ , and a fixed initial graph with average degree  $K$ . The total number of edges is preserved throughout each run.

### 3.1 Model Parameters

- **Population size ( $N$ ):** 1000.
- **Average degree ( $K$ ):** 4, 10, 40. These values control the initial connectivity of the network and the number of available social links per node.
- **Satisfaction threshold ( $p_{\text{aversion}}$ ):** 0.50, 0.75, 0.95. Each node is satisfied when at least this fraction of its neighbors are of the same type.
- **Group ratio ( $\rho$ ):** 0.5 and 0.3. The first gives a balanced 50/50 population split; the second creates a minority-majority split.
- **Initial topology:** Erdős-Rényi (ER) random graphs and Barabási-Albert (BA) scale-free graphs. ER graphs are generated with edge probability  $p_{\text{edge}} = K/(N-1)$ , and BA graphs are generated with  $m = K/2$ .
- **Iteration budget:** 5000 maximum iterations per simulation.

### 3.2 Summary of Experimental Runs

The experiment performs a full parameter sweep over:

- **Average degree ( $K$ ):** 4, 10, 40
- **Satisfaction threshold ( $p_{\text{aversion}}$ ):** 0.50, 0.75, 0.95
- **Population balance ( $\rho$ ):** 0.5, 0.3
- **Topology:** Erdős-Rényi (ER) and Barabási-Albert (BA)

This yields 36 distinct experimental conditions for the same fixed population size ( $N = 1000$ ).

Table 1: Parameter Sweep for Experiments

| Parameter                                        | Values           | Purpose                               |
|--------------------------------------------------|------------------|---------------------------------------|
| Population size ( $N$ )                          | 1000             | Fixed population size                 |
| Average degree ( $K$ )                           | 4, 10, 40        | Controls initial network density      |
| Satisfaction threshold ( $p_{\text{aversion}}$ ) | 0.50, 0.75, 0.95 | Varies tolerance requirements         |
| Group ratio ( $\rho$ )                           | 0.5, 0.3         | Balanced and minority-majority splits |
| Topology                                         | ER, BA           | Random and scale-free initial graphs  |

### 3.3 Ensemble Averaging Protocol

Each parameter combination is evaluated through 20 independent repetitions. Each repetition uses a different random seed while keeping the parameters fixed.

For every repetition, the notebook records the global homophily index at each iteration until the system reaches full satisfaction or the iteration limit is reached. Ensemble outputs are then summarized by the mean and standard deviation of each homophily trajectory.

This protocol captures the stochastic variability of the network rewiring process and supports statistically meaningful comparisons across the 36 experimental conditions.

## 4 Results and Simulation Metrics

This section details the quantitative and structural outcomes from our computational experiments. Under our algorithm, a single simulation step represents a global round where all unsatisfied nodes simultaneously evaluate their neighborhoods, dissolve one cross-group link, and establish a new link. As a result, the system exhibits rapid structural adjustments where global homophily rises sharply at the beginning of the simulation and quickly reaches a plateau.

To provide a complete architectural overview, Table 2 compiles the ensemble metrics (averaged over 20 independent repetitions) for all 36 experimental configurations executed across our parameter space.

**Important: all the numeric data mentioned in the following experiments comes from the average runs shown in Table 2.**

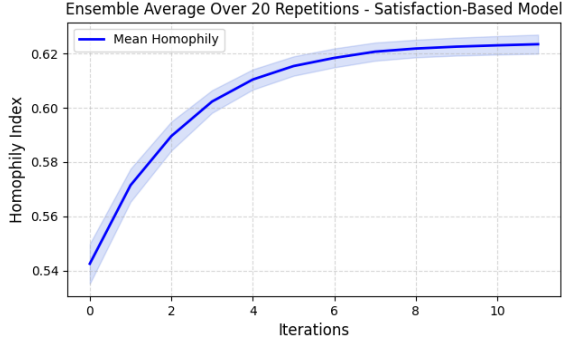
Table 2: Comprehensive Ensemble Results for All Experimental Configurations ( $N = 1000$ )

| Topology | Mean Degree ( $K$ ) | Satisfaction ( $\tau$ ) | Ratio ( $\rho$ ) | Initial $H$ | Initial Unsatisfied | Convergence Steps | Final $H$ |
|----------|---------------------|-------------------------|------------------|-------------|---------------------|-------------------|-----------|
| ER       | 4                   | 0.50                    | 0.5              | 0.5000      | 396.00              | 11.25             | 0.7160    |
| BA       | 4                   | 0.50                    | 0.5              | 0.5034      | 338.35              | 11.35             | 0.7035    |
| ER       | 4                   | 0.75                    | 0.5              | 0.4974      | 773.65              | 12.35             | 0.9471    |
| BA       | 4                   | 0.75                    | 0.5              | 0.5012      | 792.05              | 12.80             | 0.9396    |
| ER       | 4                   | 0.95                    | 0.5              | 0.4994      | 867.45              | 12.05             | 1.0000    |
| BA       | 4                   | 0.95                    | 0.5              | 0.5011      | 840.00              | 11.55             | 0.9997    |
| ER       | 10                  | 0.50                    | 0.5              | 0.4979      | 441.80              | 15.15             | 0.6246    |
| BA       | 10                  | 0.50                    | 0.5              | 0.4997      | 447.40              | 15.60             | 0.6228    |
| ER       | 10                  | 0.75                    | 0.5              | 0.5001      | 922.85              | 18.30             | 0.8650    |
| BA       | 10                  | 0.75                    | 0.5              | 0.5011      | 893.65              | 18.10             | 0.8609    |
| ER       | 10                  | 0.95                    | 0.5              | 0.4990      | 992.55              | 14.45             | 1.0000    |
| BA       | 10                  | 0.95                    | 0.5              | 0.5002      | 987.75              | 15.75             | 0.9986    |
| ER       | 40                  | 0.50                    | 0.5              | 0.5002      | 467.95              | 23.35             | 0.5554    |
| BA       | 40                  | 0.50                    | 0.5              | 0.4991      | 473.45              | 25.20             | 0.5541    |
| ER       | 40                  | 0.75                    | 0.5              | 0.5014      | 999.00              | 35.25             | 0.7966    |
| BA       | 40                  | 0.75                    | 0.5              | 0.4993      | 995.35              | 50.65             | 0.7957    |
| ER       | 40                  | 0.95                    | 0.5              | 0.5000      | 1000.00             | 34.45             | 0.9819    |
| BA       | 40                  | 0.95                    | 0.5              | 0.4996      | 1000.00             | 36.40             | 0.9817    |
| ER       | 4                   | 0.50                    | 0.3              | 0.5829      | 305.95              | 22.65             | 0.7685    |
| BA       | 4                   | 0.50                    | 0.3              | 0.5836      | 267.20              | 30.20             | 0.7693    |
| ER       | 4                   | 0.75                    | 0.3              | 0.5795      | 617.35              | 18.60             | 0.9626    |
| BA       | 4                   | 0.75                    | 0.3              | 0.5777      | 660.55              | 18.50             | 0.9624    |
| ER       | 4                   | 0.95                    | 0.3              | 0.5813      | 771.00              | 13.00             | 1.0000    |
| BA       | 4                   | 0.95                    | 0.3              | 0.5807      | 737.30              | 13.55             | 0.9999    |
| ER       | 10                  | 0.50                    | 0.3              | 0.5810      | 309.05              | 29.10             | 0.7035    |
| BA       | 10                  | 0.50                    | 0.3              | 0.5743      | 327.00              | 71.40             | 0.7080    |
| ER       | 10                  | 0.75                    | 0.3              | 0.5816      | 702.50              | 32.55             | 0.8967    |
| BA       | 10                  | 0.75                    | 0.3              | 0.5825      | 687.95              | 72.65             | 0.9040    |
| ER       | 10                  | 0.95                    | 0.3              | 0.5792      | 965.75              | 15.40             | 1.0000    |
| BA       | 10                  | 0.95                    | 0.3              | 0.5782      | 941.10              | 18.40             | 0.9994    |
| ER       | 40                  | 0.50                    | 0.3              | 0.5794      | 300.90              | 63.80             | 0.6716    |
| BA       | 40                  | 0.50                    | 0.3              | 0.5783      | 301.55              | 171.20            | 0.6775    |
| ER       | 40                  | 0.75                    | 0.3              | 0.5802      | 810.45              | 91.75             | 0.8455    |
| BA       | 40                  | 0.75                    | 0.3              | 0.5801      | 799.50              | 288.55            | 0.8563    |
| ER       | 40                  | 0.95                    | 0.3              | 0.5793      | 999.85              | 51.70             | 0.9897    |
| BA       | 40                  | 0.95                    | 0.3              | 0.5792      | 999.15              | 73.90             | 0.9916    |

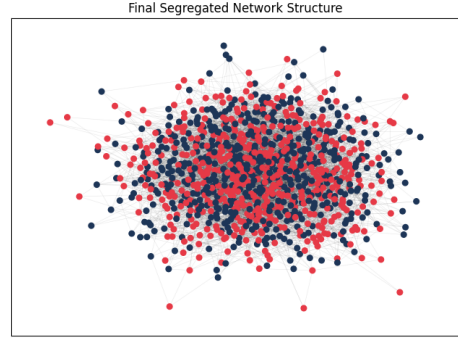
#### 4.1 Experiment 1: Effect of the Satisfaction Threshold ( $\tau$ )

This experiment analyzes how individual strictness levels modify the final graph patterns by holding the structural density and population ratio constant ( $N = 1000, K = 10, \rho = 0.5$ , Erdős-Rényi topology) while varying  $\tau$  across 0.50, 0.75, and 0.95.

When the threshold is lenient ( $\tau = 0.50$ ), the population begins with an average of 441.80 unsatisfied nodes. The system undergoes structural modifications for an average of 15.15 rounds before stabilizing. However, because agents are easily satisfied when half of their neighbors match their type, the rewiring process freezes early, leaving the final global homophily at a low value of 0.6246. The network layout in Figure 1(b) remains connected as a single structural mass where the two groups overlap significantly.



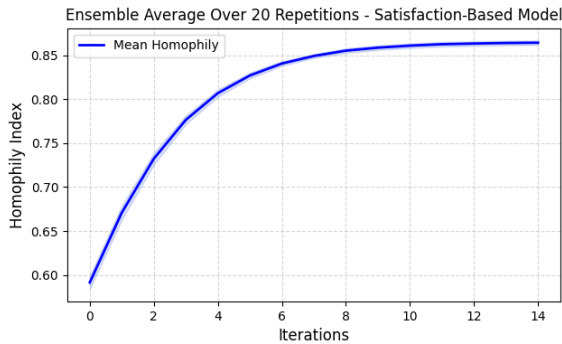
(a) Homophily Evolution ( $\tau = 0.50$ )



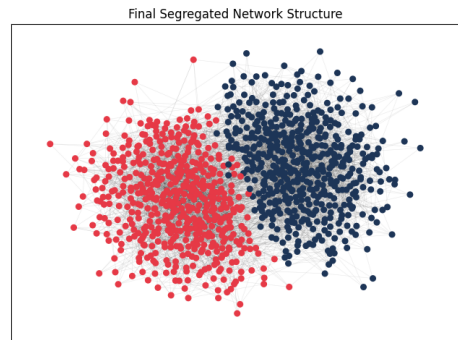
(b) Final Network Layout ( $\tau = 0.50$ )

Figure 1: System evolution under low satisfaction demands ( $\tau = 0.50, K = 10, \rho = 0.5$ , ER).

Increasing the threshold to a moderate level ( $\tau = 0.75$ ) expands the initial unsatisfied pool to 922.85 nodes. The rewiring curves rise sharply during the first 5 steps and reach a plateau around step 18.30, stabilizing at a final homophily value of 0.8650. As seen in Figure 2(b), the groups begin to clear their borders, forming distinct sub-communities that remain loosely tied by a few cross-group links.



(a) Homophily Evolution ( $\tau = 0.75$ )



(b) Final Network Layout ( $\tau = 0.75$ )

Figure 2: System evolution under moderate satisfaction demands ( $\tau = 0.75, K = 10, \rho = 0.5$ , ER).

Under strict conditions ( $\tau = 0.95$ ), nearly the entire network starts in an unsatisfied state (992.55 nodes). This strict requirement forces continuous link adjustments. The system con-

verges in 14.45 global rounds, achieving an absolute homophily index of 1.0000. Figure 3(b) captures the final structural outcome: the network cleanly breaks apart into two separate, monochromatic islands, demonstrating that high strictness demands total group isolation.

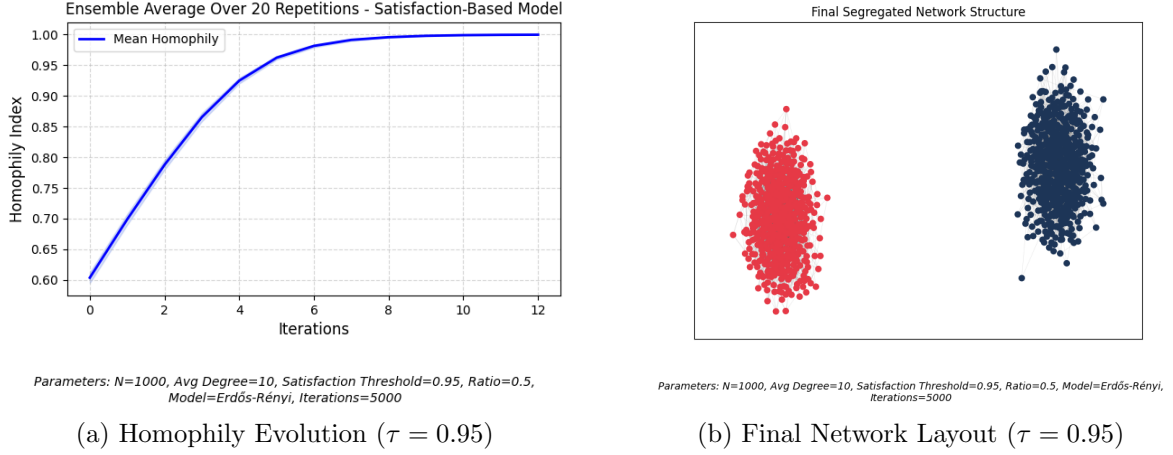


Figure 3: System evolution under strict satisfaction demands ( $\tau = 0.95, K = 10, \rho = 0.5, \text{ER}$ ).

## 4.2 Experiment 2: Effect of Network Density ( $K$ )

This experiment evaluates how the number of connections per node changes segregation dynamics by fixing  $N = 1000, \tau = 0.75, \rho = 0.5$  on Erdős-Rényi topologies, while scaling the average degree  $K$  across 4, 10, and 40.

In a sparse environment ( $K = 4$ ), nodes have few connections, making it easier to satisfy individual threshold conditions. The network contains 773.65 initial unsatisfied agents. As shown in Figure 4(a), the homophily profile rises sharply and reaches a steady state in 12.35 rounds, establishing a high final homophily index of 0.9471. The visual structure in Figure 4(b) reveals that because the network lacks structural density, the sorting process separates the groups cleanly into highly distinct local sub-graphs.

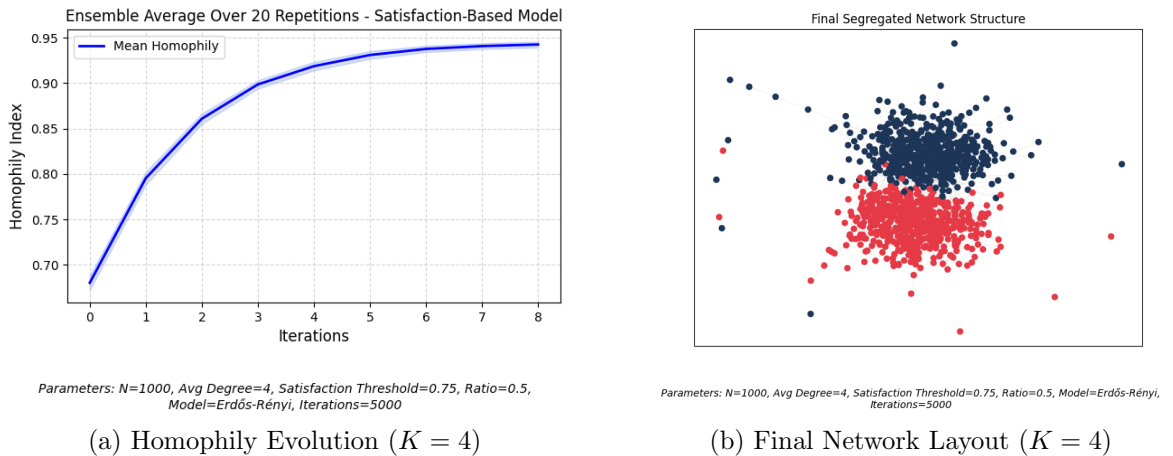
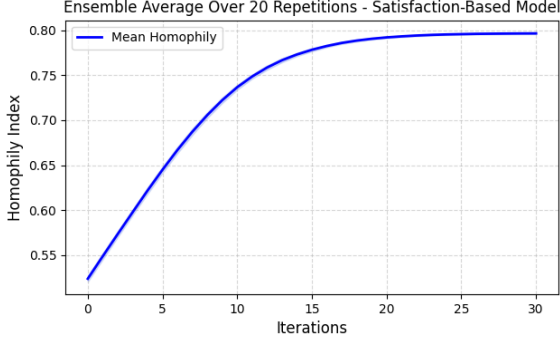


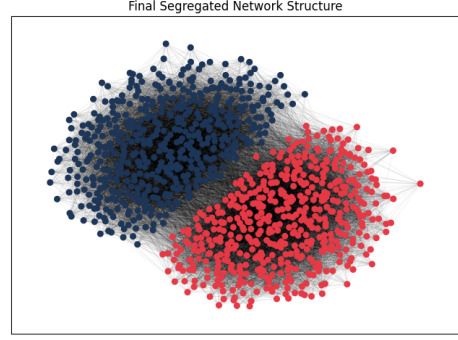
Figure 4: System evolution within a sparse structural network ( $K = 4, \tau = 0.75, \rho = 0.5, \text{ER}$ ).

When density increases to a moderate level ( $K = 10$ ), the larger number of ties makes coordination more difficult. The system requires 18.30 rounds to settle, and the final homophily index plateaus at 0.8650 (Figure 2(a)). The network layout retains a cohesive core with visible sorting, but it does not separate as cleanly as the sparse case (Figure 2(b)).

Under high structural density ( $K = 40$ ), individual nodes are deeply embedded in a dense web of connections, creating a severe structural bottleneck. Initially, 999.00 out of 1000 nodes are unsatisfied. Due to the high number of overlapping connections, the simultaneous rewiring loops struggle to resolve cross-group links efficiently. The convergence time increases to 35.25 rounds, and the final homophily curve flattens out at a low value of 0.7966 (Figure 5). The final plot illustrates this bottleneck: the network forms a heavily congested, single component where structural density physically prevents the clear separation of groups.



(a) Homophily Evolution ( $K = 40$ )

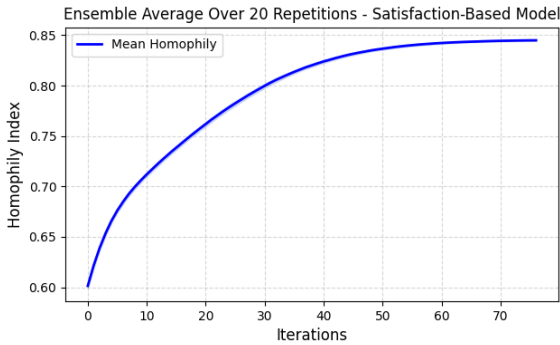


(b) Final Network Layout ( $K = 40$ )

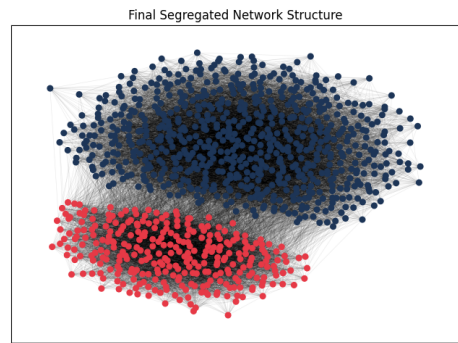
Figure 5: System evolution within a highly dense structural network ( $K = 40$ ,  $\tau = 0.75$ ,  $\rho = 0.5$ , ER).

### 4.3 Experiment 3: Impact of Network Topology (ER vs. BA)

This experiment investigates how the underlying structural layout affects the segregation process by comparing a random Erdős-Rényi (ER) graph with a scale-free Barabási-Albert (BA) topology. To highlight the structural differences, we evaluate a demanding scenario with high network density and demographic imbalance ( $N = 1000$ ,  $K = 40$ ,  $\tau = 0.75$ ,  $\rho = 0.3$ ).



(a) ER Homophily Evolution



(b) ER Final Network Layout

Figure 6: System evolution under random graph structural setups ( $K = 40$ ,  $\tau = 0.75$ ,  $\rho = 0.3$ , ER).

The empirical results show a major divergence in timeline behavior between the two models. The random ER graph starts with an initial homophily of 0.5802, contains an average of 810.45



initial unsatisfied nodes, and finishes its structural configuration in an average of 91.75 steps, reaching a final homophily index of 0.8455 (Figure 6).

In contrast, the scale-free BA graph requires an average of 288.55 steps to settle, which is more than three times longer than the ER counterpart, finishing with a final homophily of 0.8563 (Figure 7). This significant slowdown reveals a structural phenomenon that can be called the "hub bottleneck." Because BA networks are characterized by a scale-free degree distribution, a few highly-connected hub nodes hold a disproportionate number of edges. When a demographic minority ( $\rho = 0.3$ ) is introduced, a hub node requires many more sequential rounds of edge adjustments to filter out its massive pool of cross-group ties. Because the algorithm operates as a simultaneous global round, the difficulty of satisfying these highly connected hubs delays global network stability.

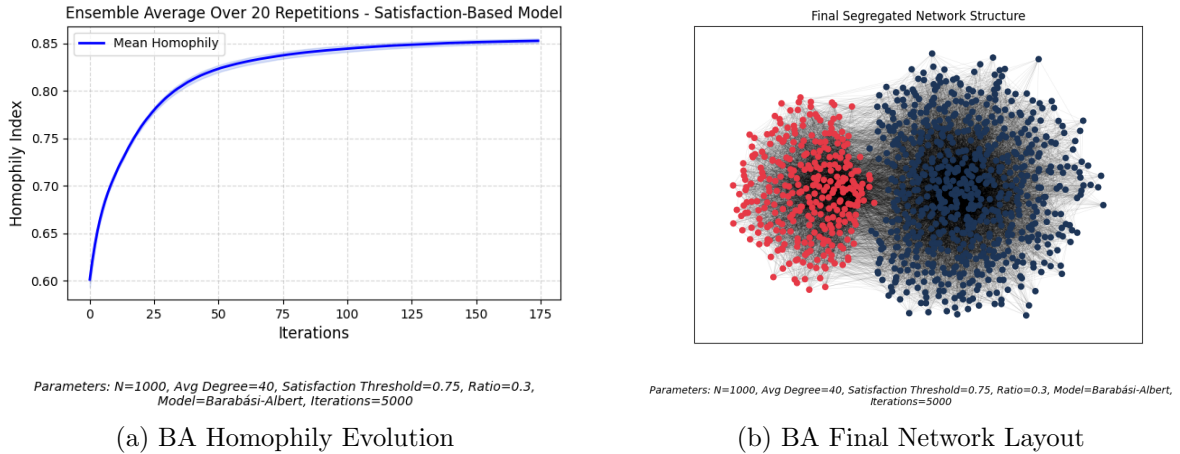


Figure 7: System evolution under scale-free hub arrangements ( $K = 40, \tau = 0.75, \rho = 0.3$ , BA).

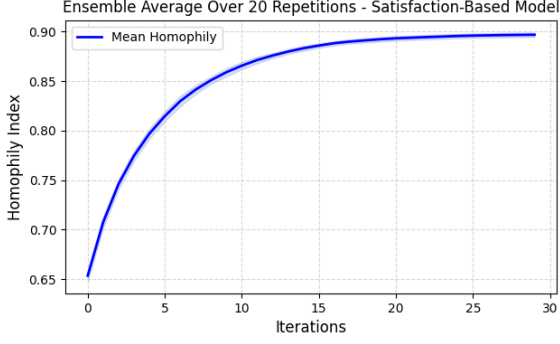
#### 4.4 Experiment 4: Effect of the Demographic Population Ratio ( $\rho$ )

This experiment examines how the balance between the two group sizes shapes the progression of network segregation. We hold the structural properties fixed ( $N = 1000, K = 10, \tau = 0.75$ , ER topology) and compare a perfectly balanced population ( $\rho = 0.5$ ) against an asymmetric minority scenario ( $\rho = 0.3$ ).

The numerical results highlight an interesting contrast between the initial setup and the total timeline of the system. The balanced population ( $\rho = 0.5$ ) begins with an initial homophily of 0.5001, has 922.85 initial unsatisfied nodes, and plateaus in 18.30 steps, reaching a final homophily index of 0.8650 (Figure 2).

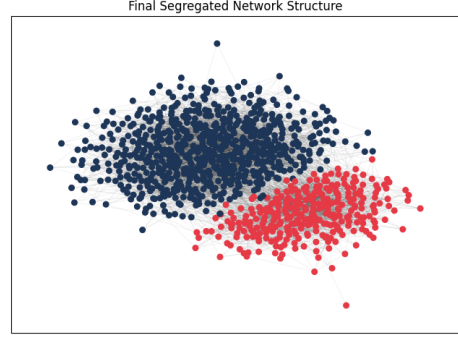
On the other hand, the asymmetric configuration ( $\rho = 0.3$ ) shows a higher initial homophily index of 0.5816, leaving a smaller pool of 702.50 initial unsatisfied nodes. This higher starting point is a mathematical product of the random graph initialization; in a network where one group represents 70% of the population, random edge placement naturally generates a higher baseline of same-group links.

However, despite this structural advantage at the start, the asymmetric network takes longer to achieve stability, requiring 32.55 steps to reach its final homophily plateau of 0.8967 (Figure 8). This behavior points to a statistical coordination challenge. Because the minority group forms a small part of the total network, unsatisfied minority agents face narrow mathematical odds when trying to randomly discover available matching partners. As a result, the simultaneous rewiring loop takes more rounds to successfully clear out the cross-group connections compared to a balanced 50/50 community.



Parameters:  $N=1000$ , Avg Degree=10, Satisfaction Threshold=0.75, Ratio=0.3, Model=Erdős-Rényi, Iterations=5000

(a) Asymmetric Profile Evolution ( $\rho = 0.3$ )



Parameters:  $N=1000$ , Avg Degree=10, Satisfaction Threshold=0.75, Ratio=0.3, Model=Erdős-Rényi, Iterations=5000

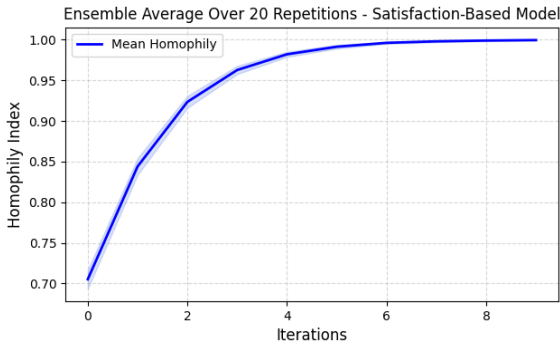
(b) Asymmetric Layout Result ( $\rho = 0.3$ )

Figure 8: System evolution under an asymmetric minority population split ( $K = 10, \tau = 0.75, \rho = 0.3$ , ER).

The visual layouts illustrate this demographic difference clearly: Figure 2(b) shows two equal-sized group patterns, while Figure 8(b) displays a smaller, tightly isolated minority cluster positioned next to a significantly larger majority network mass.

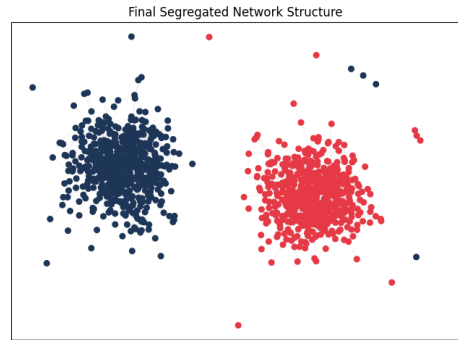
#### 4.5 Experiment 5: Perfect Segregation Limits Under High Density

This experiment explores the boundaries of achieving absolute segregation ( $H = 1.0000$ ) when individual agent preferences are extremely strict ( $\tau = 0.95$ ). We hold the structural size and population proportions constant ( $N = 1000, \rho = 0.5$ , ER topology) and compare a loosely connected graph ( $K = 4$ ) against a heavily saturated, dense graph ( $K = 40$ ).



Parameters:  $N=1000$ , Avg Degree=4, Satisfaction Threshold=0.95, Ratio=0.5, Model=Erdős-Rényi, Iterations=5000

(a) Sparse Setup Evolution ( $K = 4$ )



Parameters:  $N=1000$ , Avg Degree=4, Satisfaction Threshold=0.95, Ratio=0.5, Model=Erdős-Rényi, Iterations=5000

(b) Sparse Setup Final Layout ( $K = 4$ )

Figure 9: Achieving complete segregation under strict preferences in a sparse network ( $K = 4, \tau = 0.95, \rho = 0.5$ , ER).

The numerical results highlight a fundamental structural ceiling effect driven by edge density. In the sparse network setup ( $K = 4$ ), the population starts with an average of 867.45 unsatisfied nodes. Because each node has very few connections, finding a suitable configuration that satisfies the 95% homophily requirement is relatively straightforward. The ensemble system completes its restructuring in just 12.05 global steps, successfully reaching a perfect final homophily index of 1.0000 (Figure 9). The final visual layout in Figure 9(b) confirms that the network breaks apart into two completely isolated monochromatic islands with zero cross-group links remaining.

However, when the connectivity climbs to a dense state ( $K = 40$ ), every single node in the network initially fails the satisfaction test (1000.00 nodes unsatisfied). This extreme dissatisfaction triggers massive, parallel link swaps across the system. The simulation runs for an average of 34.45 steps but plateaus prematurely at a final homophily index of **0.9819**, failing to achieve the absolute 1.0000 limit (Figure 10).

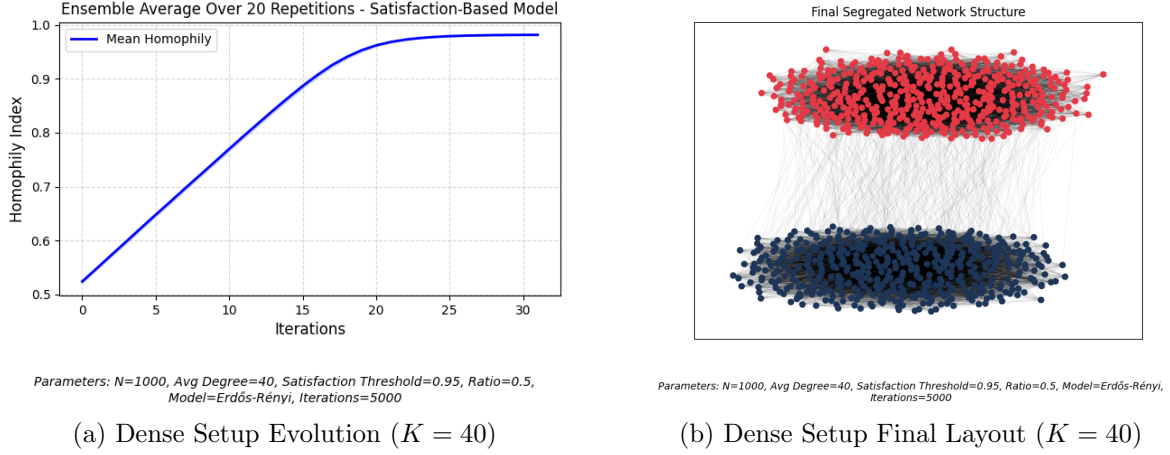


Figure 10: The structural bottleneck preventing absolute segregation in a dense network ( $K = 40$ ,  $\tau = 0.95$ ,  $\rho = 0.5$ , ER).

This outcome reveals a physical combinatorics constraint: when nodes are forced to maintain 40 connections each in a closed system, the local partner pool becomes heavily crowded. As agents simultaneously scramble to drop cross-group edges, they run into a lack of available partner slots among matching node types that they are not already connected to. This structural saturation prevents the network from completing the final microscopic adjustments needed to clear the last remaining cross-group links. The final visualization in Figure 10(b) reflects this bottleneck, showing a densely intertwined graph that cannot fracture into pure monochromatic components despite the intense pressure from individual preferences.

## 5 Discussion and Conclusion

The analysis of the simulations under the link reconfiguration model based on the simultaneous satisfaction of agents allows us to draw profound conclusions about emergent phenomena in complex networks.

### 5.1 The Satisfaction Threshold ( $\tau$ ) as an End-State Switch

The satisfaction threshold  $\tau$  determines whether the network stays integrated or breaks apart. When preferences are tolerant ( $\tau = 0.50$ ), the system freezes quickly with low global homophily ( $H \approx 0.62$ ). Because agents are happy if only half their neighbors match their type, the rewiring stops early, keeping both groups mixed within a single network mass.

In contrast, a strict threshold ( $\tau = 0.95$ ) forces continuous changes. The system rewires until all cross-group links are gone, achieving in some cases perfect homophily ( $H = 1.0000$ ). Visually, this strictness forces the network to split into two completely isolated monochromatic islands.

## 5.2 Link Density ( $K$ ) and the Combinatorial Bottleneck

Network density (average degree  $K$ ) acts as a physical constraint on how fast groups can sort themselves. In sparse networks ( $K = 4$ ), nodes have very few links to manage. The sorting process is highly efficient, reaching a high homophily plateau (0.9471) in about 12 steps.

In dense networks ( $K = 40$ ), a severe bottleneck occurs. Because each node must maintain 40 connections, the local search space becomes crowded. Nodes compete simultaneously for the same limited pool of matching partners. This structural saturation slows convergence to 35.25 steps and prevents the system from achieving perfect segregation, leaving final homophily stagnant at 0.7966 in same conditions.

## 5.3 Network Topology and the "Hub Bottleneck"

Comparing random networks (Erdős-Rényi) with scale-free networks (Barabási-Albert) reveals a "hub bottleneck." While both topologies behave similarly in balanced setups, they diverge drastically in dense, asymmetric conditions ( $K = 40, \rho = 0.3$ ). Here, the BA network takes over three times longer to stabilize (288.55 steps) than the ER network (91.75 steps).

Because hubs have a massive number of connections, filtering out cross-group links when their own group is a minority requires many rounds of updates. Since global stability requires every single node to be satisfied, these hyper-connected hubs act as structural traps that delay the entire system.

## 5.4 Demographic Population Ratio ( $\rho$ ) and Search Asymmetry

The demographic ratio  $\rho$  changes the baseline conditions of the simulation. Reducing one group to a minority ( $\rho = 0.3$ ) creates a higher random baseline homophily ( $H \approx 0.58$ ) at the start due to basic probability.

However, despite this head-start, asymmetric networks take longer to reach a steady state (32.55 steps vs. 18.30 steps for balanced networks). Minority nodes face a statistical challenge: finding available, matching partners in a graph dominated by the majority requires more collective iterations, slowing down the system's evolutionary speed.

## 5.5 Conclusion and Future Work

In conclusion, this project successfully models Schelling's segregation dynamics on complex networks. Our main findings show that:

1. The individual tolerance threshold controls the final network layout, forcing a complete split into isolated components when preferences are strict.
2. High link density creates a physical bottleneck that limits perfect sorting due to local search space saturation.
3. Scale-free hubs heavily delay stability when combined with a demographic minority.

## References

- [1] Henry, A. D., Pralat, P., & Zhang, C.-Q. (2011). Emergence of segregation in evolving social networks. *Proceedings of the National Academy of Sciences*, 108(21), 8605-8610.
- [2] Schelling, T. C. (1971). Dynamic models of segregation. *Journal of Mathematical Sociology*, 1(2), 143-186.